

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

MSN LABORATORIES PRIVATE LTD. and MSN
PHARMACEUTICALS INC.,¹
Petitioners,

v.

BAUSCH HEALTH IRELAND LIMITED,
Patent Owner.

IPR2022-00722
Patent 7,041,786 B2

Before JEFFREY N. FREDMAN, CYNTHIA M. HARDMAN, and
MICHAEL A. VALEK, *Administrative Patent Judges*.

VALEK, *Administrative Patent Judge*.

JUDGMENT

Final Written Decision on Remand
Determining No Challenged Claims Unpatentable
35 U.S.C. §§ 144, 318
Granting Patent Owner's Motion to Exclude
37 C.F.R. § 42.64(c)

¹ MSN Laboratories Private LTD. and MSN Pharmaceuticals Inc. (collectively "MSN" or "Petitioner") filed a motion for joinder and a petition in IPR2023-00016 and have been joined as a petitioner in this proceeding.

I. INTRODUCTION

This case is on remand following the appeal of our prior final written decision (Paper 78, “Decision”). In the Decision, we found that none of the challenged claims 1–6 of U.S. Patent No. 7,041,786 B2 (Ex. 1001, “the ’786 patent”) had been shown unpatentable because while Petitioner² had established a *prima facie* showing of obviousness, that showing was “outweighed by Patent Owner’s comparatively strong evidence demonstrating unexpected results.” Dec. 60.

Those findings and our conclusion on obviousness were appealed to the Court of Appeals for the Federal Circuit (“Federal Circuit”) See Paper 83 (“CAFC Decision”). The Federal Circuit did not disturb our determination that Petitioner “established a *prima facie* case of obviousness.” See CAFC Dec. 6–8. However, it took issue with the unexpected results analysis in the Decision, explaining that “without more explanation from the Board as to its determination that the experiments [offered by Patent Owner to show unexpected results] compared the challenged claims to the closest prior art, we cannot reach a judgment on MSN’s challenges to the Board’s holding of non-obviousness.” *Id.* at 14. Accordingly, the Federal Circuit vacated the Decision and remanded the case for further consideration of the unexpected results evidence. *Id.*

² The Petition (Paper 1, “Petition”) in this proceeding was filed by Mylan Pharmaceuticals Inc. (“Mylan”). MSN filed a substantively-identical petition and was joined as a petitioner in this proceeding. MSN assumed an understudy role to Mylan who filed the Petitioner Reply (Paper 38, “Reply”) to Patent Owner’s Response (Paper 70, “Resp.”). See IPR2023-00016, Paper 14 (agreeing to understudy role). Mylan settled while the case was on appeal and was subsequently terminated as a party upon remand to the Board. Paper 89, 3. MSN has continued as the sole Petitioner on remand. *Id.*

More specifically, the Federal Circuit identified “three reasons” why “the Board’s determination that the[se] experiments compared the claimed invention with the closest prior art lack[ed] adequate explanation” and instructed us to address these issues on remand. *See* CAFC Dec. 11–13. First, the Court stated we “should determine, based on the evidence put forward by the parties, whether plecanatide exhibits topoimerism and whether the testing of the challenged claims should encompass both the active and inactive forms.” *Id.* at 12.³ Second, the Court instructed us to reconsider what the closest prior art compound is in view of Patent Owner’s “contention that highly pure A-form uroganylin could be maintained as a lyophilized powder,” noting that “[i]f it is true . . . that uroganylin can be maintained and therefore tested in its active A form as a lyophilized powder, that would support that the experiments [which were conducted in solution] did not use the closest prior art.” *Id.* Third, the Court observed that the Decision did not sufficiently address “whether the experiments submitted by [Patent Owner] actually used” the closest prior art and instructed the Board “to reevaluate the experimental evidence consistent” with the determinations noted above. *Id.* at 12–13.

³ Plecanatide is the name of the peptide recited in claim 1. It is a synthetic analog of another peptide, human uroganylin. CAFC Dec. 3. The only difference in the amino acid sequence of these peptides is that the aspartic acid residue at position 3 in uroganylin has been substituted with a glutamic acid in plecanatide. *Id.* In the present context, topoimerism refers to the capability of a peptide having the same amino acid sequence to “exist in different three-dimensional structures with different levels of biological activity,” e.g., an active “A-Form” and inactive “B-Form.” *Id.* at 3; *see also* Ex. 2024 ¶¶ 50–51 (“[A] ‘topoisomer’ is the same chemical compounds having different spatial arrangement of the various atoms in the molecule that occur because of rotations around bonds.”).

On remand, both parties sought authorization to file post-remand briefing addressing the three issues identified in the CAFC Decision. *See* Ex. 3004, 1–2 (joint email from counsel).⁴ Patent Owner also asked to submit additional evidence regarding the same. *Id.* Following a telephonic hearing, we authorized post-remand briefing and certain additional evidence from Patent Owner “limited to facts regarding the same experiments and data already of record and presented in Patent Owner’s Response.” Paper 92, 3–5; *see also* Ex. 3006, 1. We gave Petitioner an opportunity to respond with additional evidence of its own. Paper 92, 5–6.

Pursuant to that authorization, Patent Owner filed an opening brief⁵ (Paper 94, “PO Remand Brief”) with additional exhibits, including declarations from Dr. Shailubhai (Ex. 2072) and Dr. Pennington (Ex. 2023). Petitioner filed a responsive brief (Paper 98, “Pet. Remand Resp.”) with additional exhibits, including a declaration from Dr. Peterson (Ex. 1078) and Patent Owner filed a reply (Paper 103, “PO Remand Reply”). Depositions were also taken of each of these witnesses. *See* Ex. 1079 (Shailubhai); Ex. 1080 (Pennington); Ex. 2077 (Peterson).

After considering the parties’ arguments and evidence, in view of the guidance in the CAFC Decision, we again find that Petitioner has not shown by a preponderance of the evidence that the challenged claims of the ’786 patent are unpatentable.

⁴ Exhibit 3004 was filed as Exhibit 3005 in P-TACTS.

⁵ Because Patent Owner bears the burden of production on unexpected results, the parties proposed, and we agreed, that Patent Owner should file the opening brief on remand. *See* Paper 92, 2.

II. BACKGROUND

A. Real Parties in Interest

Petitioner identifies MSN Laboratories Private Limited and MSN Pharmaceuticals, Inc. as real parties in interest (“RPI”). Paper 91, 2. Patent Owner identifies itself and Salix Pharmaceuticals, Inc. as RPI. Paper 4, 2.

B. The ’786 Patent

The ’786 patent issued on May 9, 2006, from a utility application filed on March 28, 2002, and claims priority to a provisional application filed on January 17, 2002. Ex. 1001, codes (22), (45), (60).

The ’786 patent is titled “Guanylate Cyclase Receptor Agonists for the Treatment of Tissue Inflammation and Carcinogenesis.” Ex. 1001, code (54). According to the Specification,

[t]he present invention relates to the therapeutic use of guanylate cyclase receptor agonists as a means for enhancing the intracellular production of cGMP. The agonists may be used either alone or in combination with inhibitors of cGMP-specific phosphodiesterase to prevent or treat cancerous, pre-cancerous and metastatic growths, particularly in the gastrointestinal tract and lungs. In addition, the agonists may be used in the treatment of inflammatory disorders such as ulcerative colitis and asthma.

Id. at 1:14–23.

The Specification explains that “[u]roguanylin, guanylin and bacterial ST [i.e., heat-stable enterotoxin] peptides are structurally related peptides that bind to a guanylate cyclase receptor and stimulate intracellular production of cyclic guanosine monophosphate (cGMP).” Ex. 1001, 1:26–29. “This results in the activation of the cystic fibrosis transmembrane conductance regulator (CFTR),” which in turn “leads to stimulation of

sodium and water secretion into the intestinal lumen.” *Id.* at 1:29–36.

“Therefore, by serving as paracrine regulators of CFTR activity, cGMP receptor agonists,” such as uroguanylin, guanylin, and bacterial ST peptides, “regulate fluid and electrolyte transport in the GI tract.” *Id.* at 1:36–39.

“The present invention is based upon the development of new agonists of guanylate cyclase receptor” that “are analogs of uroguanylin.” Ex. 1001, 3:3–6. The Specification describes the sequence for its “most preferred” uroguanylin analog in SEQ ID NO:20, which is reproduced below.

Asn¹ Asp² Glu³ Cys⁴ Glu⁵ Leu⁶ Cys⁷ Val⁸ Asn⁹ Val¹⁰ Ala¹¹ Cys¹² Thr¹³ Gly¹⁴ Cys¹⁵ Leu¹⁶

Id. at 5:7–12. SEQ ID NO:20 provides the primary structure of a 16-amino acid long peptide. The sequence in SEQ ID NO:20 differs from that of the “Parent compound: uroguanylin” (shown in SEQ ID NO: 1) in that it has a “Glu³,” i.e., a glutamic acid (Glu) at position 3 in the peptide sequence, whereas human uroguanylin has an “Asp³,” i.e., an aspartic acid (Asp) at that position.⁶ *See id.* at 11 (Table 2). This is the only difference between the sequence in SEQ ID NO:20 and the naturally-occurring human uroguanylin sequence in SEQ ID NO:1. *Id.*; *see also* Ex. 1002 ¶¶ 24–25.

According to the Specification, the peptide of SEQ ID NO:20 has been found to have enhanced biological activity as an agonist of cGMP production due to its enhanced binding constant for the guanylate cyclase receptor, and is superior to uroguanylin with regard to temperature and protease stability

⁶ Both parties use a superscript to denote the position of the residue in an amino acid sequence. *See, e.g.*, Pet. 1, 38; Resp. 2 (referring to “Glu³” and “Asp³”). In addition to Glu, other synonyms for glutamic acid appearing in the record include “glutamate” and the letter “E” from the single-letter code for amino acids. In addition to Asp, other synonyms for aspartic acid include “aspartate” and the letter “D.”

and with regard to its biological activity at the physiologically favorable pH range (pH 6 to 7) in the large intestine.

Ex. 1001, 5:17–23.

Table 4 in the Specification shows data from a “T84 cell-based assay for determining the intracellular levels of cGMP.” Ex. 1001, 15:35–36. “Peptide analogs were custom synthesized by Multiple Peptide Systems, San Diego, Calif., and by Princeton Biomolecules, Langhorne, Pa. Biological activity of the synthetic peptides was assayed as previously reported.” *Id.* at 15:36–40 (citing the Shailubhai publication).⁷ In addition, the ’786 patent states that the “Peptides shown in Table 4 were custom synthesized and purified (>95% purity) using a published procedure.” *Id.* at 15:53–54 (citing Klodt).⁸ Table 4 is reproduced below and shows data for human uroguanylin (SP301) and plecanatide (SP304).

⁷ K. Shailubhai et al., *Uroguanylin Treatment Suppresses Polyp Formation in the Apc^{Min/+} Mouse and Induces Apoptosis in Human Colon Adenocarcinoma Cells via Cyclic GMP*, 60 *Cancer Research* 5151–5157 (2000) (Ex. 2075) (“the Shailubhai publication”).

⁸ J. Klodt et. al., *Synthesis, biological activity and isomerism of guanylate cyclase C-activating peptides guanylin and uroguanylin*, 50 *J. Peptide Res.* 222–230 (1997) (Ex. 2020) (“Klodt”).

TABLE 4

Peptide agonists evaluated for biological activity in the T84 cell bioassay.		
SEQ ID NO.*	Compound Code	cGMP Level** (pmol/well)
1	SP301	205
6	SP302	225
7	SP303	195
20	SP304	315
14	SP306	0
4	SP310	0
21	SP316	275

*SEQ ID's for SP301, SP304 and SP316 are the precise amino acid sequences for these analogs as given in the text.

**Intracellular cGMP level observed in T84 cells following treatment with 1 micromolar solution of the respective peptide agonist for 30 minutes. The value observed for SP304 was statistically significant with a $p > 0.5$.

“As shown in Table 4 [above], SP304 (SEQ ID NO:20) gave the greatest enhancement of intracellular cGMP of all the” peptides tested. *Id.* at 15:53–58.

C. Challenged Claims

The Petition challenges claims 1–6. Claims 1–3 and 6 are independent claims. They provide as follows:

1. A peptide consisting of the amino acid sequence of SEQ ID NO:20.
2. A composition in unit dose comprising a guanylate cyclase receptor agonist peptide consisting of the amino acid sequence of SEQ ID NO:20.
3. A composition in unit dose form comprising:
 - a) a guanylate cyclase receptor agonist peptide consisting of the amino acid sequence of SEQ ID NO: 20; and
 - b) at least one compound selected from the group consisting of: a cGMP-dependent phosphodiesterase inhibitor, an anti-inflammatory agent, an antiviral agent and an anticancer agent.

6. A peptide conjugate comprising polyethylene glycol (PEG) attached to a peptide consisting of the amino acid sequence SEQ ID NO:20.

Ex. 1001, 37:2–12, 38:7–9. Claims 4 and 5 additionally limit the compositions of either claim 2 or 3 to a “dose form . . . selected from the group consisting of a tablet, a capsule, a solution and an inhalation formulation” (claim 4) and to further comprise “one or more excipients” (claim 5). *Id.* at 38:1–6.

Claim 1 is directed to a peptide with the sequence of SEQ ID NO:20. The parties’ papers and exhibits refer to this peptide by several names, including [Glu³]-human uroguanylin, [Glu³]-UG, plecanatide, and SP-304. *See, e.g.*, Pet. 5 (noting that Patent Owner “markets [Glu³]-human uroguanylin as TRULANCE[®] (plecanatide)”; Resp. 1; Ex. 1001, 10:42–63 (referring to the Glu³-substituted analog of human uroguanylin as SP304). Plecanatide “is the active pharmaceutical ingredient in Trulance[®], which is FDA-approved for the treatment of chronic idiopathic constipation . . . and irritable bowel syndrome with constipation.” Resp. 1; *see also* Pet. 5. (similar); Ex. 1055 (prescribing information for “TRULANCE (plecanatide) tablets”).

D. Asserted Grounds of Unpatentability

Petitioner asserts the following grounds of unpatentability:

Claim(s) Challenged	35 U.S.C. §⁹	Reference(s)/Basis
1	103(a)	Currie ¹⁰ , Li ¹¹
2, 4, 5	103(a)	Currie, Li, Narayani ¹²
3–5	103(a)	Currie, Li, Narayani, Campieri ¹³
6	103(a)	Currie, Li, Ekwuribe ¹⁴

E. Incorporation of Prior Decision

All of Petitioner’s asserted grounds of unpatentability rely on the same combination of Currie and Li for the limitations of claim 1 and the similar limitations recited in the other independent claims. In the Decision, we found that, based on the teachings in those references, an ordinarily-skilled artisan would have been motivated both to select human uroguanylin as a lead compound and modify its sequence by making a conservative Glu³ substitution for uroguanylin’s Asp³. Dec. 15–37. We did not, however, credit Petitioner’s other purported rationales for that modification. *Id.* at 34–37. Those findings were not disturbed on appeal. *See* CAFC Dec. 6–8 (rejecting

⁹ The Leahy-Smith America Invents Act, Pub. L. No. 112–29, 125 Stat. 284 (2011) (“AIA”), included revisions to 35 U.S.C. § 103 that became effective after the filing of the application that led to the ’786 patent. Therefore, we apply the pre-AIA version of 35 U.S.C. § 103.

¹⁰ U.S. Pat. 5,489,670, issued Feb. 6, 1996 (Ex. 1005) (“Currie”).

¹¹ Z. Li et al., *Purification, cDNA Sequence, and Tissue Distribution of Rat Uroguanylin*, 68 *Regulatory Peptides* 45–56 (1997) (Ex. 1006) (“Li”).

¹² R. Narayani et al., *Polymer-coated Gelatin Capsules as Oral Delivery Devices and their Gastrointestinal Tract Behavior in Humans*, 7 *J. Biomater. Sci. Polymer Edn.* 39–48 (1995) (Ex. 1007) (“Narayani”).

¹³ M. Campieri et al., *Oral Budesonide is as Effective as Oral Prednisolone in Active Crohn’s Disease*, 41 *Gut* 209–214 (1997) (Ex. 1008) (“Campieri”).

¹⁴ U.S. Pat. 5,359,030, issued Oct. 25, 1994 (Ex. 1009) (“Ekwuribe”).

Patent Owner’s alternative arguments for affirmance). We incorporate by reference the portions of the Decision addressing Petitioner’s *prima facie* showing herein. *See* Dec. 8–37 (§§ III.A–D and E.i–ii). We consider the findings and explanation therein to the extent it bears on our analysis below.

III. ANALYSIS

The focus on remand is Patent Owner’s unexpected results showing. In the Decision, we found there were three categories of unexpected results data that evidenced non-obviousness: (1) interconversion stability, (2) potency, and (3) binding affinity.¹⁵ *See* Dec. 37–59. Categories (1) and (2) are based on the Patent Owner’s own data in Table 4 of the Specification

¹⁵ Patent Owner previously asserted a fourth category of unexpected results—increased heat stability, but we found that its data did not demonstrate a sufficient “difference in kind” as compared to uroguanylin. Dec. 53–56. That finding was not appealed (*see* CAFC Dec. 9, n. 1) and we do not address that category further here.

and studies described in Exhibits 2027 and 2028.¹⁶ *Id.* at 38. Category (3) is based on data that was published in a research article, Liu.¹⁷ *Id.* at 56.

On remand, Patent Owner continues to assert that these three categories of data demonstrate unexpected results compared to the closest prior art on remand. *See* PO Remand Br. 2. Petitioner disputes the same. *See generally* Pet. Remand Resp. (urging “Patent Owner has not met its burden to show its alleged ‘unexpected results’ compared the invention to the closest prior art, or that the unexpected results are commensurate in scope with the claims.”).

We begin by addressing the three issues identified by the Federal Circuit. *See* CAFC Dec. 12–13. After addressing those issues, we assess the specifics of each category of Patent Owner’s results as well as Petitioner’s argument that those results are not reasonably commensurate with the scope of the claims. Finally, we weigh the record as a whole to reach a conclusion on obviousness.

¹⁶ Exhibits 2027 and 2028 are reports for Study No. SP-PH-001 and Study No. SP-PH-004 respectively. These reports were prepared at Dr. Shailubhai’s direction and approved by him in his capacity as an executive at Synergy Pharmaceuticals, the original assignee of the ’786 patent. Ex. 2066 ¶¶ 30–34; *see also* Ex. 2027, TRUL00018205; Ex. 2028, TRUL00018262 (signature page). They were considered by Synergy’s management as “part of the decision-making process for plecanatide” and submitted to the FDA as part of the New Drug Application (“NDA”) for Trulance®. Ex. 2072 ¶ 13; *see also* Paper 61, 9 (Patent Owner representing that “Preclinical Study Report Nos. SP-PH-001 (Ex. 2027) and SP-PH-004 (Ex. 2028) are a part of the NDA submitted to the FDA for Trulance®”); Paper 26, 1 (similar).

¹⁷ D. Liu et. al., *In Vivo Imaging of Human Colorectal Cancer Using Radiolabeled Analogs of the Uroguanylin Peptide Hormone*, 29 *Anticancer Research* 3777–3784 (2009) (Ex. 2046) (“Liu”).

A. Whether plecanatide exhibits topoisomerism

The parties agree that plecanatide, like uroguanylin, exhibits topoisomerism, meaning it “exists in, and can interconvert between, two topoisomeric forms, one active and the other inactive.” PO Remand Br. 1, 4 (citing Ex. 2024 ¶¶ 50, 53; Ex. 2040); *see also* Pet. Remand Resp. 28 (acknowledging same). This means that claim 1, which recites a “peptide consisting of the amino acid sequence of SEQ ID NO:20,” encompasses that peptide regardless of its topoisomeric form. Ex. 1063 ¶ 150 (“Sequence ID No. 20 does not require any particular topoisomer . . . but instead would encompass a composition of peptide consisting of Sequence ID No. 20 that is purely an ‘active’ topoisomer, purely an ‘inactive’ topoisomer, or any mixture of two or more topoisomers sharing Sequence ID No. 20.”).

Patent Owner agrees that “Claim 1 encompasses plecanatide in both topoisomeric forms,” but attempts to distinguish claim 2, urging that it “focuses on the active topoisomer of plecanatide by reciting ‘a *guanylate cyclase receptor agonist* peptide consisting of the amino acid sequence of SEQ ID NO:20.” PO Remand Br. 5 (emphasis in original). Petitioner responds, asserting that “[t]o the extent . . . Patent Owner is now raising claim construction to argue that claim 2 does not stand or fall with claim 1,” something it has never done before, “it is far too late . . . for such gamesmanship.” Pet. Remand Resp. 29. Petitioner further points to evidence suggesting that even the inactive topoisomer is likely to have some agonist activity. *Id.* at 28 (citing Ex. 1078 ¶¶ 22–23).

We agree that it is too late for Patent Owner to argue that claim 2 does not stand or fall with claim 1. As Petitioner points out, “Patent Owner has never separately argued the patentability of claims 2–6” before. Pet. Remand

Resp. 28 n. 16. And even now, Patent Owner does not articulate any argument for treating claim 2 (or any of the other challenged claims) differently from claim 1. PO Remand Br. 5 (asserting only that claim 2 “focuses on plecanatide’s active form”); *see generally* PO Reply (electing not to respond to Petitioner’s arguments regarding claim 2). But even if we were to consider such an argument, we do not discern any reason to treat claim 2 differently than claim 1 in assessing Patent Owner’s unexpected results showing. It is undisputed that the active topoisomer of plecanatide is a guanylate cyclase receptor agonist peptide and the evidence of record shows such. *See infra* § III.C.ii–iii (showing plecanatide’s activity and binding affinity). Thus, there is a nexus between the peptide of claim 2 and the alleged unexpected results.

Moreover, to the extent Patent Owner contends that claim 2 is limited to only the active topoisomer, we disagree. First, Patent Owner does not cite evidence to support its assertion that the “inactive form of plecanatide is not an agonist for the claimed receptor.” *See* PO Remand Br. 5. In contrast, Petitioner points to evidence that even the “inactive” topoisomer of uroguanylin shows some guanylate cyclase receptor agonist activity in cGMP assays. Ex. 1078 ¶ 23; Ex. 2010, 239 (Fig. 8); Ex. 2074, 48 (observing that the inactive topoisomer “showed maximal 10%” activity of the active topoisomer). For this reason, we credit the testimony of Petitioner’s expert Dr. Peterson that an ordinarily-skilled artisan would *not* expect the inactive form of plecanatide to be “wholly inactive” or “devoid of agonist activity.” Ex. 1078 ¶ 23.

Second, as Petitioner points out, nothing in the claims or Specification of the ’786 patent defines the level of activity required for the peptide to be

considered a “guanylate cyclase receptor agonist.” Pet. Remand Resp. 29: Ex. 1078 ¶ 24. Accordingly, even if the “inactive” topoisomer exhibits greatly reduced activity relative to the “active” form, we do not read claim 2 to exclude the former by reciting a “guanylate cyclase receptor agonist peptide.” Ex. 1001, 37:4–5.

As for the Federal Circuit’s related question about whether the testing should encompass both of plecanatide’s topoisomers (*see* CAFC Dec. 12), we note that it likely does even if the parties’ evidence does not directly show such. Because plecanatide is subject to topoisomerism it exists as a mixture of those topoisomers when it is in solution that trends to an equilibrium state of the isomeric forms. Ex. 2024 ¶¶ 51, 73. As explained below, the relevant assays are carried out in solution. *See infra* § III.B. Thus, even when testing plecanatide that had been purified to a high degree of its active form, it is likely some minute amount of inactive topoisomer was present in these tests, either as an impurity or due to interconversion during testing. *Id.* But, as explained below in § III.D, we do not understand this to undermine the weight of Patent Owner’s unexpected results evidence.

B. What is the closest prior art

In the Decision, we found that the closest prior art is “human uroguanylin as it naturally exists, i.e., a mixture with some interconversion between topoisomers.” *See* Dec. 39. That finding was premised on evidence demonstrating that uroguanylin readily interconverts between topoisomers in solution. *See id.* at 38–39 (citing Ex. 2010, 238; Ex. 2011, 28, 31; Ex. 2024 ¶¶ 67–68; Ex. 2025 ¶¶ 49–52).

The Federal Circuit took issue with our finding because it “did not address MSN’s contention that highly pure A-form uroguanylin could be maintained as a lyophilized powder.” CAFC Dec. 12. It explained that:

If it is true, as MSN contends, that uroguanylin can be maintained and therefore tested in its active A form as a lyophilized powder, that would support MSN’s argument that the experiments did not use the closest prior art. But in finding that the closest prior art is a “mixture” of A- and B-form uroguanylin, the Board only cited evidence stating that uroguanylin solutions could not be maintained for extended periods of time. Its broad conclusion that a mixture of A and B-form uroguanylin—regardless of its formulation—is the closest prior art therefore does not reasonably follow from the solution-specific evidence it cited in support.

Id. (citation omitted). Taking the Court’s guidance into account, we now reconsider the arguments and evidence on this point.

First, it is undisputed that human uroguanylin is subject to topoimerism. *See generally* PO Remand Br.; Pet. Remand Resp.; *see also* Ex. 2011, 27; Ex. 2020, 223; Ex. 2024 ¶¶ 64–67. Because uroguanylin is subject to topoimerism, it naturally exists as a mixture of its topoimeric forms. Ex. 2024 ¶¶ 51, 73. This is particularly the case when it is in solution at biologically relevant temperature and pH. Ex. 2024 ¶ 68; Ex. 2025 ¶¶ 49–51; *see also* Ex. 2010, 236 (“At 37°C, 25% of both uroguanylin-15 isomers are interconverted within 24 h.”); Ex. 2011, 29 (Fig. 5 showing more rapid interconversion at 37°C and acidic pH). Under such conditions, it was understood that uroguanylin readily interconverts between its topoimeric forms. *Id.*

It was also known that Reversed-phase High Performance Liquid Chromatography (“RP-HPLC”) could be used to isolate and purify each of

the topoisomeric forms of human uroguanylin to at least 95% purity.

Ex. 2011, 29 (reporting that each uroguanylin topoisomer “could be separately isolated at purities greater than 99%” as determined by RP-HPLC); Ex. 2074, 48 (same); Ex. 2075, 5153 (using RP-HPLC to purify the biologically active form to “~99% purity” for use in T84 bioassay); *see also* Ex. 2020, 224, 229 (RP-HPLC to purify followed by T84 bioassay to characterize active form).

In addition, it is not disputed that the closest prior art is human uroguanylin that has been purified to at least 95% of the active topoisomer (“purified (>95%)”). Pet. Remand Resp. 5, 29 (“the commonly accepted purity criteria for a peptide to be used in research is greater than 95% purity”); Ex. 1078 ¶ 25; Ex. 1080, 35:24–36:10 (both sides’ declarants agreeing); *see also* PO Remand Br. 5–7, 9 (arguing that the purified (>95%) active form was used in its experiments). But importantly, this does not mean that the closest prior art contains only the active topoisomer of uroguanylin. Because human uroguanylin is “a dynamic peptide,” even purified (>95%) active form human uroguanylin can include a small amount (<5%) of the inactive topoisomer. *See, e.g.*, PO Remand Br. 5, 8 (“Though human uroguanylin’s topoisomers can be isolated and separately purified, they are still prone to interconversion.”), 16 (“Some amount of the inactive topoisomeric form is inevitable.”); Pet. Remand Resp. 9 (“The parties also agree that, even after purification, uroguanylin was known to interconvert from its active form to an inactive form).

Moreover, while the record shows that the purified active form of uroguanylin can be lyophilized to a powder form and refrigerated to stabilize it against interconversion, these are not the relevant conditions for its testing.

See Ex. 2010, 236 (“No conversion of the lyophilized uroguanylin isomers after long term storage of up to 3 months at -20°C ”); Ex. 2011, 29 (purity “maintained for more than one year when [purified] compound was stored in a freezer as an amorphous powder”). The experiments Patent Owner relies upon to show unexpected results are liquid assays. Ex. 2072 ¶ 11 (“[T]he T84 cell bioassay is a solution-based test.”). So too are the assays described for related peptides in the prior art.¹⁸ Ex. 2010, 232 (T84 cell bioassay in solution); Ex. 2011, 28 (same); Ex. 2020, 224 (same); Ex. 2075, 5152 (same); *see also* Ex. 2027, TRUL00018211; Ex. 2028, TRUL00018268; Ex. 2046, 3778 (solution-based cell binding assays and pharmacokinetic studies). Such assays are designed to mimic biological conditions. See Ex. 2072 ¶ 11 (“[B]ecause human uroguanylin rapidly goes into solution in the gastrointestinal tract, and because human uroguanylin needs to be in solution for it to be biologically active in the gastrointestinal tract, we tested it in well-known solution-based assays to acquire information about how it might behave *in vivo*.”). For these reasons, a POSA would understand that, even when starting from a highly isomerically pure powder, some interconversion may occur after the powder is solubilized for testing. See Ex. 2011, 31 (observing that “interconversion may occur during course of the analysis” and results may vary “if peptide solutions have been left at room temperature for a few days, especially under acidic conditions”).

¹⁸ To the extent Petitioner argued on appeal “that uroguanylin can be maintained *and therefore tested* in its active A form *as a lyophilized powder*” (*see* CAFC Dec. 12 (emphasis added)), it does not maintain that position in its post-remand briefing. Nor does the record show that any relevant bioassay can be conducted on these peptides in their solid form.

C. Whether Patent Owner's experiments used the closest prior art

Patent Owner contends that the comparative experiments described in the Specification and Exhibits 2027 and 2028 “tested ‘best on best,’ that is human uroguanylin’s purified active form versus stable, interconversion-resistant plecanatide to demonstrate the claimed invention’s unexpected results.” PO Remand Br. 2.¹⁹ According to Patent Owner, “[t]he declarations of Drs. Kunwar Shailubhai and Michael Pennington reliably establish that synthesized human uroguanylin purified to 95% or more of the active topoisomer was used and tested in the study reported in the ’786 example and in the studies of Ex. 2027 and Ex. 2028.” *Id.* at 3. Patent Owner argues that their testimony is corroborated by citations to Klodt, the Shailubhai publication, and other references in the ’786 patent and Exhibits 2027 and 2028—all of which consistently teach that RP-HPLC is used to purify uroguanylin to its active topoisomer for use in the T84 bioassay and other testing. *See id.* at 3, 10–14 (citing evidence).

Petitioner disputes Patent Owner’s “assertion of a ‘best on best’ comparison.” Pet. Remand Resp. 2. According to Petitioner, Patent Owner’s current position that these experiments tested purified (i.e., >95%) active form plecanatide against purified active form uroguanylin contradicts what it previously told the European Patent Office (EPO).²⁰ *Id.* at 5–6. According to

¹⁹ We address the parties’ arguments concerning whether the experiments in Liu also used the closest prior art separately in § III.C.iii.

²⁰ A European counterpart to the ’786 patent was previously the subject of an opposition proceeding before the EPO. *See* Ex. 1067 (excerpts from EPO file history). We previously granted Patent Owner’s motion to exclude portions of Exhibit 1067, i.e., letter briefs and an experimental report submitted by the opponent in the EPO opposition, because those materials

Petitioner, “relying on this *exact same evidence* in the proceedings before the EPO, Patent Owner argued the opposite—that the claimed invention was compared to ‘uroguanylin in its naturally-occurring form (i.e., *a mixture of [active] and [inactive] forms*)[.]” *Id.* at 5 (quoting Ex. 1067, 103) (alterations in original). Therefore, Petitioner contends, “Patent Owner was either being untruthful then or is being untruthful now.” *Id.* at 6.

Petitioner further contends that the testimony of Drs. Shailubhai and Pennington is uncorroborated and that these witnesses lack personal knowledge or recollection of critical aspects of the comparative experiments. *See* Pet. Remand Resp. 10–21. According to Petitioner, the documents and other references relied on by Patent Owner to corroborate their testimony fails to corroborate because this evidence lacks details about how the uroguanylin was shipped, stored, and prepared for use in those studies. *Id.* at 22–26.

Patent Owner replies that its “statements before the EPO are not inconsistent with having testified purified active uroguanylin” because it “never argued that plecanatide was tested against a 1:1 mixture of uroguanylin topoisomers or against anything other than its purified (>95%) active form” and was merely responding to the opponent’s argument that its data were “obtained using the A-form of uroguanylin *only*.” PO Remand Reply 10 (citing Ex. 1067, 100, 102, 103) (quotations omitted). As for Petitioner’s lack of corroboration arguments, Patent Owner argues they fail

were hearsay without an applicable exception. *See* Dec. 61–73. We distinguished the excluded portions of Exhibit 1067 from the letter briefs submitted by Patent Owner’s counsel at Mintz Levin in the EPO opposition (Ex. 1067, 84–86, 99–103, 138–140). Dec. at 63–64.

to consider the totality of the record and misapply the rule of reason analysis that considers the evidence “as a whole, not individually.” *Id.* at 3–4 (quoting *NFC Tech., LLC v. Matal*, 871 F.3d 1367, 1371–72 (Fed. Cir. 2017)).

We find that Patent Owner has the better position because its evidence shows that the results in the ’786 patent and Exhibits 2027 and 2028—which Patent Owner uses to support its claims of unexpectedly better interconversion stability and potency—were obtained from “best on best” testing of the purified (>95%) active topoisomers of human uroguanylin and plecanatide.²¹ Dr. Shailubhai, one of the named inventors, testified that his team used synthesized human uroguanylin that was purified to 95% or more of the active topoisomer in these experiments. Ex. 2072 ¶¶ 4–5. To do this, Dr. Shailubhai had uroguanylin prepared by manufacturers who purified it to at least 95% of the active topoisomer using RP-HPLC, according to published procedures in Exhibit 2020 (“Klodt”), Exhibit 2074 (“Chino”), and Ex. 2075 (“the Shailubhai publication”). *Id.* ¶¶ 6–9. Dr. Shailubhai explains that the purified uroguanylin was shipped to him as a lyophilized

²¹ Our analysis focuses on the evidence and arguments as it relates to the uroguanylin used in these experiments because Petitioner does not dispute Patent Owner’s position that the purified (>95%) active topoisomer of plecanatide was used therein. Pet. Remand Resp. 8, n. 3 (“MSN and Patent Owner have both consistently represented that *plecanatide* was purified relative to its topoisomer”); *see also* Ex. 1001, 15:54–55 (all “[p]eptides shown in Table 4 were . . . purified (>95% purity)”). But even if the plecanatide had not been purified to remove the inactive topoisomer prior to testing, the presence of a greater amount of plecanatide’s inactive topoisomer in the samples that were tested would only serve to strengthen the unexpected nature of the superior results obtained for plecanatide. *See infra* § III.C.i–iii.

powder and stored at or below -20 °C until the time of testing. *Id.* ¶¶ 10, 12. As explained above, that frozen lyophilized powder uroguanylin does not readily interconvert. Ex. 2010, 236; Ex. 2011, 29. When the assay was ready, a solution of uroguanylin was prepared “and immediately used . . . in the assay” to minimize interconversion as much as possible. *Id.*

Dr. Shailubhai’s testimony is corroborated by the testimony from Dr. Pennington. Dr. Pennington worked at Bachem Biosciences, Inc. (“Bachem”), the company that supplied the test peptides for the studies described in Exhibits 2027 and 2028. Ex. 2073 ¶ 6; *see also* Ex. 2027, TRUL00018210; Ex. 2028, TRUL00018267 (identifying Bachem as the peptide supplier). Dr. Pennington testified that the human uroguanylin supplied to Dr. Shailubhai during the timeframe of these studies came from him and that he purified it using RP-HPLC such that it contained greater than 95% of the active topoisomer. Ex. 2073 ¶¶ 8–10. Dr. Pennington also corroborated Dr. Shailubhai’s testimony that the uroguanylin was shipped to him as a lyophilized powder kept at or below -20 °C. *Id.* ¶ 11.

The references cited in the ’786 patent and Exhibits 2027 and 2028 further corroborate Dr. Shailubhai’s testimony that the studies described in Table 4 of the Specification and in Exhibits 2027 and 2028 used human uroguanylin that had been purified (>95%) of the active topoisomer. The ’786 patent states that the T84 bioassay in its example “was assayed as previously reported” in the Shailubhai publication. Ex. 1001, 15:36–41, 17:32 (citing Ex. 2075). Exhibits 2027 and 2028 similarly state that the T84 bioassays were conducted “using [the] published procedure” in the

Shailubhai publication. Ex. 2027, TRUL00018211;²² Ex. 2028, TRUL00018268. The Shailubhai publication makes clear that its T84 bioassay procedures involve testing of uroguanylin peptides that had been purified to remove the inactive topoisomer. Ex. 2075, 5153 (“The major peptide peak, exhibiting a potent biological assay, was further purified by reverse phase-HPLC to ~99% purity and used for this study.”).

In addition to citing the Shailubhai publication, the ’786 patent states that the assayed peptides were “custom synthesized and purified (>95% purity) using a published procedure” in Klodt. Ex. 1001, 15:53–54 (citing Ex. 2020). According to Patent Owner’s expert, one of ordinary skill in the art would understand this to mean 95% or greater isomeric purity, i.e., no more than 5 percent of anything else, including topoisomers of the same peptide. Ex. 1060, 114:19–116:10. That testimony is supported by the disclosure in Klodt, which refers to Chino’s observation that solid-phase synthesis of human uroguanylin “yields two topological isomers which could be separated by HPLC.” Ex. 2020, 223 (citing Ex. 2074).²³ Klodt further describes procedures for synthesis, purification, and analytical HPLC methods that it uses to “confirm” Chino’s observation of uroguanylin’s two

²² We understand the citation to endnote “(8)” in Exhibit 2027 to be a typographical error because the Shailubhai publication is the only reference in the endnotes. *See* Ex. 2027, TRUL00018214.

²³ The parties and witnesses refer to two different references authored by Chino—Exhibit 2011 and Exhibit 2074. Both Chino references describe human uroguanylin as having an active (compound I) and inactive (compound II) topoisomer and methods for separating the two by RP-HPLC. *See, e.g.*, Ex. 2011, 27, 29 (“isolated at purities greater than 99%” using RP-HPLC); Ex. 2074, 45 (Summary), 48 (“purity > 99% on RP-HPLC”). Exhibit 2074 is the “Chino” cited in Klodt. Exhibit 2011 is cited in Ex. 2028 (TRUL00018272) and elsewhere.

isoforms. *See id.* 223–224 (describing synthetic procedures and RP-HPLC methods); 227 (confirming Chino’s result).²⁴ Indeed, Petitioner’s expert, Dr. Peterson, appears to agree that the procedure taught in Klodt will yield topoisomerically–pure human uroguanylin. *See* Ex. 1063 ¶ 37 (testifying that Klodt teaches how to separate and characterize uroguanylin’s active topoisomer).

Nevertheless, on appeal, Petitioner took what the Federal Circuit observed were “contradictory positions as to what level of purity Klodt refers.” CAFC Dec. 13. That is, in one context Petitioner urged that “Klodt refers to purity with respect to other compounds, not topoisomerism,” whereas in another it urged that the “procedure in Klodt was used to purify peptides at the topoisomer level.” *Id.* Based on the evidence above, we find that the record supports the latter, topoisomeric-purity interpretation. Indeed, in its post-remand brief Petitioner does not dispute that the procedure taught in Klodt will result in peptides purified to greater than 95% of their active form. Instead, Petitioner attempts to defend against the charge that it took contradictory positions on Klodt on appeal by urging that it has “consistently represented that *plecanatide* was purified relative to its topoisomer.” Pet. Remand Resp. 8 n. 3. The problem with that argument is that it implies the ’786 patent’s statement that the peptides in Table 4 were “purified (>95%

²⁴ Klodt refers to “uroguanylin-(97-112),” which is the 16-amino acid long peptide referred to as human uroguanylin in this proceeding. *Compare* Ex. 2020, 226 (Table 1, SEQ ID No. 14) with Ex. 1001, 11 (Table 2, “Parent compound: uroguanylin”). Klodt separately refers to a longer peptide “uroguanylin-(89-112)” having eight more amino acids (*see* Ex. 2020, 226 (SEQ ID No. 15)) that neither side contends is relevant to the closest prior art analysis.

purity)” means that some of those peptides (plecanatide) were purified at the topoisomeric level, but others (uroguanylin) were not. Ex. 1001, 15:53–54. We see no basis in the record for adopting Petitioner’s inconsistent, peptide-specific interpretation of that statement.

Accordingly, the fact that the ’786 patent and Exhibits 2027 and 2028 expressly state that the experiments therein use the procedures in the Shailubhai publication and Klodt further corroborates Dr. Shailubhai’s testimony that these experiments used the purified (>95%) active form of human uroguanylin. Moreover, the fact those procedures are cited in the ’786 patent itself demonstrates corroboration in a contemporaneous document because the patent was filed near in time to the study dates indicated in Exhibits 2027 and 2028. *Compare* Ex. 1001 (March 2002 filing date), *with* Ex. 2027, TRUL00018203; Ex. 2028, TRUL0018260 (September to December 2001 study dates).

Two other documents further corroborate aspects of Patent Owner’s testimonial evidence. The first is Exhibit 2076, which is an “Analytical Data Sheet” dated May 24, 2012, for an order of uroguanylin supplied to Dr. Shailubhai from Peptides International. Ex. 2076, 1, 3. It includes an “HPLC profile” for the uroguanylin in that order showing 96.1% purity and indicates it was being provided as a powder and stored at -20 °C. *Id.* at 1–3. Dr. Pennington explained that Peptides International was the company he joined in 2010, after leaving Bachem, and that in that new role he continued to synthesize and supply Dr. Shailubhai with human uroguanylin purified to remove the inactive topoisomer. Ex. 2073 ¶ 7. The data sheet in Exhibit 2076 is dated about a decade after the experiments Patent Owner relies upon to show unexpected results. However, Dr. Pennington testified that Exhibit

2075 confirms “his usual specifications for synthesized human uroguanylin purified to greater than 95%” and “delivered [to Dr. Shailubhai] as a lyophilized powder at or below -20 °C” as he “always did.” *Id.* ¶¶ 12–15. Accordingly, while Exhibit 2076 is not a data sheet for the particular peptides used in the experiments in Exhibits 2027 and 2028 and therefore not direct evidence that those peptides were purified to their active topoisomer, it corroborates Dr. Pennington’s testimony regarding the specifications he says he has always used to supply human uroguanylin in a document that long predates any of the parties’ present disputes.

The second document is Exhibit 2040—a letter from Dr. Pennington to Dr. Shailubhai dated March 31, 2004, which quotes a price for a “scale up” in the production of SP-304 (plecanatide) as well as a price for “the parent compound uroguanylin.” Ex. 2040, 1. The letter notes that uroguanylin is “notorious” for interconversion and difficult to purify, whereas plecanatide had been found to be less so based on Bachem’s initial work and therefore “[p]urification was greatly simplified allowing us to offer the very favorable price [for plecanatide] above.” *See id.* (quoting a price for plecanatide that is 5 time less than uroguanylin on a per mg basis). As Dr. Pennington testified, his letter in Exhibit 2040 confirms that Dr. Shailubhai “regularly requested that [he] provide him with solid-phase synthesized human uroguanylin and SP-304 (plecanatide) purified with respect to their active topoisomers.” Ex. 2073 ¶ 12; *see also* Ex. 2072 ¶ 6 n. 3 (similar testimony from Dr. Shailubhai regarding Exhibit 2040). While this letter is dated a few years after the experiments in Exhibits 2027 and 2028, it refers back to the “same priority treatment as you [Dr. Shailubhai] have already experienced with us.” Ex. 2040, 1. Accordingly, the letter in

Exhibit 2040 further corroborates Dr. Pennington's testimony that he "always" supplied uroguanylin, purified to its active topoisomer, to Dr. Shailubhai. Ex. 2027 ¶ 12.

Petitioner's arguments disputing Patent Owner's showing are not persuasive. *See* Pet. Remand Resp. 5–26. Petitioner's strongest argument is the allegedly contradictory statements from Patent Owner's counsel in the EPO opposition. *Id.* at 5–8. Those statements apparently refer to the same experimental data that Patent Owner relies on here, but Patent Owner's counsel before the EPO argued the data "were obtained using uroguanylin in its naturally-occurring form (i.e., a mixture of A- and B-forms) and [plecanatide]. This is in contrast to the . . . data presented by Opponent 1, which specifically uses the A-form of each peptide." Ex. 1067, 103.

Patent Owner's attempt to reconcile its EPO statements with its present position is not compelling. *See* PO Remand Br. 15–16; PO Remand Reply 10. While it is true, as Patent Owner asserts, that it "never argued that plecanatide was tested against a 1:1 mixture of uroguanylin topoisomers" and even a purified (>95%) active form uroguanylin "is a mixture" of topoisomer, Patent Owner's statements to the EPO suggest something different than its present position, i.e., that the uroguanylin was tested as a naturally-occurring mixture of topoisomers as opposed to uroguanylin that had been purified by RP-HPLC to at least 95% of the active topoisomer. *See* PO Remand Reply 10.

However, even accepting the contradictory import of the EPO statements, those statements do not overcome the weight of Patent Owner's present showing that its experiments, in fact, tested human uroguanylin that had been purified to 95% or greater of the active topoisomer. As explained

above, Patent Owner’s present showing is premised on testimony from multiple witnesses and corroborated by contemporaneous citation to experimental methods from multiple references—all of which consistently teach the testing of topoisomerically pure uroguanylin—as well as additional, ancillary documents (Exhibits 2040 and 2076). While the EPO statements undercut that showing to some extent, they do not overcome the weight of the evidence. Put differently, Petitioner frames this as a binary issue—either Patent Owner’s explanation of what was tested was correct “then” or it is correct “now.” *See* Pet. Remand Resp. 6. We find that the current record, which is aided by more fulsome evidence and argument regarding that testing than that present in the EPO record (Ex. 1067), demonstrates that Patent Owner’s present explanation is the correct one.

Petitioner’s arguments that Patent Owner’s witnesses lack personal knowledge are unavailing. *See* Pet. Remand Resp. 9–16. Petitioner argues that Dr. Shailubhai did not observe the synthesis and purification of the tested uroguanylin and did not personally perform any the comparative studies. *Id.* at 10–12 (asserting the studies were conducted by Dr. Dheer’s team and recorded in Dr. Dheer’s notebook). But this does not show that Dr. Shailubhai lacked personal knowledge of the facts in his declaration.

To the contrary, Dr. Shailubhai’s testimony that he supervised the team that performed the experiments in the ’786 patent and Exhibits 2027 and 2028 is un rebutted. Ex. 2072 ¶ 4; *see also* Ex. 2023 ¶¶ 10, 17, 30; Ex. 2066 ¶¶ 10, 17, 30 (“I directed the preparation and testing” and “have first-hand knowledge” of the test data). That team included Dr. Dheer who was the principal scientist conducting those experiments. Ex. 2072 ¶ 13; Ex. 1079, 42:22–43:8. While he did not perform the experiments, Dr. Shailubhai

testified that he: (1) authorized Dr. Dheer to perform them; (2) “personally observed” and was “actually present in the lab when the experiments were run;” (3) was “involved in reviewing and looking at the methods to be used” in those experiments; and (4) “supervised and reviewed all the data.” Ex. 1079, 42:22–44:2. This is more than sufficient to establish foundation for his testimony describing how those experiments were conducted and what materials were used.

Moreover, while Dr. Shailubhai did not personally observe the synthesis and purification of the tested peptides, which were prepared by third party manufacturers such as Bachem, he did instruct them to provide topoisomerically-pure peptides according to the procedures in Klodt and Chino. Ex. 2072 ¶ 6; *see also* Ex. 1079, 27:7–12 (manufacturers given methods and publications); 61:16–62:217 (“it was our instructions to all of those companies to use those publications [Klodt and Chino] for matters of synthesis and purification”), 91:14–17 (same for Bachem). Dr. Pennington’s testimony confirms that he received that instruction and consistently followed it for the uroguanylin supplied by Bachem. Ex. 2073 ¶¶ 8–12. There is no reason to suspect that the peptide manufacturers identified in the Specification did not similarly follow the instructions and methods that Dr. Shailubhai gave them to prepare these peptides.

Petitioner speculates that Bachem and Dr. Pennington may not have actually supplied the peptides used in the studies recorded in Exhibits 2027 and 2028. *See* Pet. Remand Resp. 14–16 (arguing that the Specification, which identifies different manufacturers for the peptides in Table 4, describes the same experiments in Exhibits 2027 and 2028 and assuming therefore that Bachem did not provide the uroguanylin for any of the

experiments Patent Owner relies upon). We are not persuaded. Exhibits 2027 and 2028 both identify Bachem as the peptide supplier. Ex. 2027, TRUL00018210; Ex. 2028, TRUL00018267. And Dr. Pennington testified that he provided uroguanylin to Dr. Shailubhai in the relevant time frame. Ex. 2073 ¶ 9.²⁵ But even if Petitioner’s speculation that Exhibits 2027 and 2028 describe the same experiments as the ’786 patent were correct, this would not undermine the thrust of Dr. Shailubhai’s testimony that these experiments (whether separate or part of the same set of assays) tested uroguanylin that had been purified to 95% or greater of the active topoisomer.

Petitioner additionally contends that, even if “the uroguanylin peptides in the studies were purified to remove the inactive topoisomer, none of these experiments speak to what happened *after* the uroguanylin peptides were purified, i.e., how the uroguanylin was shipped, stored and prepared for use in the studies.” Pet. Remand Resp. 21–22. We disagree. Both Dr. Shailubhai and Dr. Pennington offered testimony within their personal knowledge on these subjects. *See* Ex. 2073 ¶¶ 11–12 (testifying that following purification the purified uroguanylin was stored and shipped as a stable lyophilized powder at temperatures at or below -20 °C); Ex. 2072 ¶ 10 (testifying that the received lyophilized powder “was immediately stored at or below -20 °C in my lab” and “[o]nce an assay was ready, we prepared a solution of human uroguanylin from the stabilized lyophilized powder and

²⁵ Patent Owner also identifies differences in the experimental data and Dr. Peterson’s comparisons of the data in the ’786 patent and Exhibit 2027 that tend to undermine Petitioner’s position that data arises from the same experiment. *See* PO Remand Reply 7–8, n. 2.

immediately used it in the assay”). Accordingly, Petitioner’s argument is not that there is *no* evidence on these points, but instead that some of the details in the witness testimony is not also recorded in the presently-available documentary evidence.

Looking at the evidence as a whole, we find that the lack of documentary support for some aspects of the witness testimony does not undermine Patent Owner’s showing. First, as explained above, the documentary evidence corroborates the key fact that Dr. Shailubhai requested and received peptides for use in these experiments that had been purified according to prior art methods to 95% or greater of the active topoisomer. *See, e.g.*, Ex. 1001, 15:39–41, 53–54 (peptides were synthesized, purified, and assayed according to the methods in the Shailubhai publication and Klodt); Ex. 2020, 223–224 (RP-HPLC method to isolate and purify human uroguanylin); Ex. 2027, TRUL00018211; Ex. 2028, TRUL00018268 (assays performed as described in the Shailubhai publication); Ex. 2040 (quoting prices for purified plecanatide and uroguanylin); Ex. 2075 (T84 assay method using uroguanylin purified by RP-HPLC to isolate the active topoisomer at ~99% purity).

Second, any details missing from the documentary record (e.g., that peptides were refrigerated prior to use and that mixed solutions were promptly tested) are sufficiently filled by credible fact witness testimony. Indeed, as Patent Owner observes, “[i]t defies logic to suggest that skilled artisans, including Drs. Shailubhai and Pennington, were aware of uroguanylin’s propensity to interconvert but that Dr. Shailubhai would have obtained the purified active form of uroguanylin—at great cost—only to

ship, store, and prepare it in a manner permitting its interconversion.” PO Remand Reply 9.

Third, that there are some gaps in documents and witness memory is not unreasonable given that the experiments in question occurred almost 25 years ago. As Dr. Shailubhai explains, the experiments were conducted at Synergy Pharmaceuticals toward the end of 2001. Ex. 2073 ¶ 13. Synergy Pharmaceuticals went through bankruptcy proceedings before being purchased by the present Patent Owner. *Id.* Moreover, Dr. Dheer, whose lab notebooks are referenced in Exhibits 2027 and 2028, is now deceased. *Id.* It is therefore “not surprising” that Patent Owner was unable to locate some of the potentially relevant documents. *See NFC Tech.*, 871 F.3d 1367, 1374 (explaining that adequate corroboration for inventor testimony can be found even on somewhat limited documentation, particularly for events that occurred twenty or thirty years prior where “the inventor’s inability to submit [certain] documents [i]s ‘not surprising’ given the amount of time that had passed”) (quoting *Loral Fairchild Corp. v. Matsushita Elec.*, 266 F.3d 1358, 1365 (Fed. Cir. 2002)).²⁶ We find the available documentary and testimonial (Pennington) evidence adequate, particularly when considering the passage of time in this case, to corroborate Dr. Shailubhai’s testimony that the purified (>95%) active topoisomer of uroguanylin was used in the experiments described in the ’786 patent and Exhibits 2027 and 2028.

²⁶ These cases address the rule of reason analysis for corroboration of inventor testimony of conception. Both sides agree that a “rule of reason” analysis should apply to assess the credibility of Dr. Shailubhai’s testimony here. Pet. Remand Resp. 18–19; *see* PO Remand Reply 1–3 (“under the proper application of the rule of reason, the evidence as a whole corroborates [Dr. Shailubhai’s] testimony”).

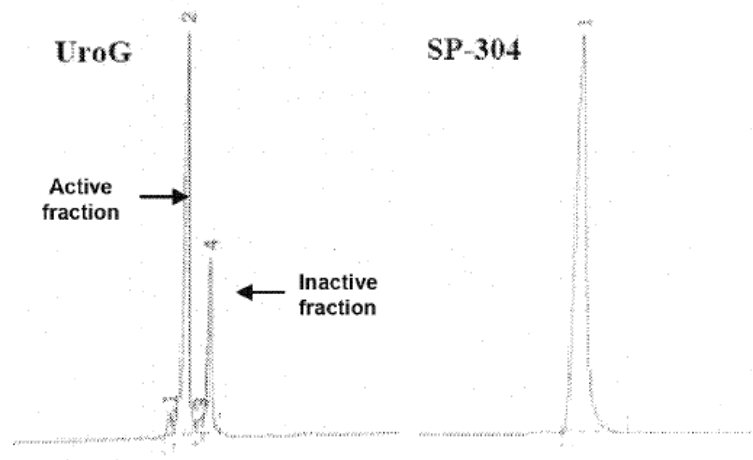
In sum, we find Patent Owner's evidence that its experiments involved "best on best" testing of the claimed invention against the closest prior art credible. Having considered and addressed the issues identified in the CAFC Decision, we now turn to the specifics of Patent Owner's evidence and the parties' arguments for each category of results.²⁷

i. Interconversion stability

Patent Owner argues that "[p]lecanatide's particular combination of structural features yielded a peptide that is unexpectedly stabilized against interconversion [e.g., from its active to its inactive form] as compared to human uroguanylin." Resp. 58. As evidence, Patent Owner offers data from an experiment involving "[s]amples of plecanatide (SP-304) and human uroguanylin (UroG)." *Id.* (citing Ex. 2028, TRUL00018269). As explained above, these peptides had been purified to greater than 95% of the active topoisomer prior to testing. The samples were "incubated for 16 hours at 37°C in a solution having a pH of 3.0" and "then analyzed by HPLC." *Id.* (citing Ex. 2028, TRUL00018269). The results of that analysis are shown in the figure reproduced below. Ex. 2028, TRUL00018269.

²⁷ Much of the analysis below repeats the findings in our prior Decision, which were not addressed in the CAFC Decision.

Figure 1: HPLC Chromatographs of Uroguanylin and SP-304 Following 16-Hour Incubations at pH 3.0 in Aqueous Media at 37°C



The figure above shows HPLC chromatographs of uroguanylin on the left and SP-304 on the right. The uroguanylin chromatograph shows a first peak labeled “Active fraction” next to a second, smaller peak labeled “Inactive fraction.” *Id.* In contrast, the SP-304 chromatograph shows only a single peak. According to Patent Owner, the result shown in this data “would not have been expected, for nothing in the art would have suggested that the Asp³ to Glu³ substitution would have improved human uroguanylin’s isomeric stability, let alone to this degree.” Resp. 59 (citing Ex. 2024 ¶¶ 210–215).

We find Patent Owner’s evidence persuasive to show that the Glu³ substitution results in an unexpected improvement in interconversion stability. As explained above, RP-HPLC was a known technique for separating the topoisomers of human uroguanylin. Ex. 2010, 230–231; Ex. 2011, 27–28; Ex. 2020, 223; Ex. 2028, TRUL00018269; *see also* Ex. 1063 ¶¶ 37, 40 (Petitioner’s expert confirming same). It was also known that one of human uroguanylin’s topoisomers was biologically active,

whereas the other, i.e., what we and the parties refer to as the “inactive” topoisomer, had significantly less or no activity at the same concentration. Ex. 2011, 31 (“compound I shows significant activity in the cGMP producing assay, whereas compound II is practically inactive”); *see also* Ex. 2010, 238–239 (Fig. 8) (showing the “A isomer” had a potency “with a minimal concentration between 10^{-7} and 10^{-8} M,” but that the B isomer “had virtually no effect on cGMP level at 10^{-6} M”); Ex. 2074, 48 (compound II “showed maximal 10% of the activity of compound I at 10^{-6} M”). The chromatograph for uroguanylin in Patent Owner’s data is consistent with these teachings, showing two peaks representing the active and inactive topoisomers. *Compare* Ex. 2028, TRUL00018269 (Fig. 1) *with* Ex. 2011, 28 (Fig. 2). The chromatograph for SP-304 shows a markedly different result, i.e., a “single peak,” which was confirmed “using the cGMP T84 cell assay” to represent “the fully active conformer of SP-304.” Ex. 2028, TRUL00018269; Ex. 2024 ¶¶ 208–209.

We agree with Patent Owner and credit Dr. Davies’s testimony (Ex. 2024 ¶¶ 210–215) that a POSA would not have expected the Glu³ substitution to produce such a result. Indeed, the premise for Petitioner’s “conservative substitution” rationale is that the Glu³ substitution would have been expected to produce an analog having similar properties to human uroguanylin. *See, e.g.*, Pet. 1, 36 (stating that a POSA “would have expected this particular conservative substitution to retain excellent receptor-activating activity”); Reply 3 (urging that the reason to modify a known compound may be an expectation that the new compound will have “similar properties to the old”) (internal quotations omitted). Patent Owner’s HPLC data, however, demonstrate a difference in kind between the data for SP-304

(a single HPLC peak) and uroguanylin (two distinct peaks corresponding to the active and inactive topoisomer) after incubation under the same conditions. As Dr. Davies explains, “topoisomers exist as a mixture of compounds with different[] spatial conformations” that are “continuously interconverting between spatial conformations” with “varying degrees of interconversion stability.” *See* Ex. 2024 ¶¶ 51–56. Thus, Patent Owner’s data show that the Glu³ substitution results in a peptide that is demonstrably more stable against interconversion than the parent, unmodified peptide.

Exhibit 2010 (“Marx”) further shows that this result would not have been expected. The Asp³ residue in human uroguanylin is the third residue from the amino-terminal end of that peptide. But Marx teaches that “isomerization [e.g., interconversion to the other topoisomer] is not affected by the amino-terminal region of uroguanylin.” Ex. 2010, 236. Marx also suggests that a different modification, i.e., adding a “third disulfide bond apparently would lead to a preference of a structure similar to the” biologically active isomer. *See* Resp. 39–40 (quoting Ex. 2010, 235). Thus, Marx suggests that a POSA seeking to stabilize uroguanylin against interconversion to the inactive isomer should pursue modifications other than the Glu³ substitution.

Nevertheless, Petitioner cites Marx and Klodt, urging that the “prior art suggested acidic residues at positions 2-3 contribute to stabilization.” Reply 28. We disagree. As explained above, Marx teaches that it is the “ionizable side-chains” in Asp² and Asp³ that “may be involved in the control of stabilization of the two isomers.” Ex. 2010, 236. The substitution of one acidic residue (Asp) with another residue having an acidic side chain (Glu) does not affect this; both the Glu³ and Asp³ peptide would still have an

ionizable side chain at the same position. Petitioner also refers to Klodt's observation that "the 'sterical bulk' of 'other residues' far from the C-terminus may influence interconversion." Reply 13 (quoting Ex. 2020, 228). But that teaching does not identify Asp³. Rather, what Klodt says is that "other residues, *in particular between the inner cysteine residues*, also influence [stability of topoisomers] by their sterical bulk." Ex. 2020, 228 (emphasis added). Asp³ is not located between the cysteine residues in human uroguanylin. *See* Ex. 1005, 1:50–55 (SEQ ID NO: 1); Ex. 1006, 52 (Fig. 6a) (showing that human uroguanylin has cysteine residues at positions 4, 7, 12, and 15). Accordingly, neither Marx nor Klodt supports Petitioner's position that a POSA would have expected the result shown in Patent Owner's HPLC data.

Petitioner's argument that Patent Owner has not shown that the incubation conditions for its experiment "(e.g., pH 3 for 16 hours)" are "relevant" is unavailing. Reply 27. First, Petitioner does not explain what it objects to regarding these incubation conditions. *See id.* To the extent Petitioner contends the conditions differ from those *in vivo* when a peptide is orally administered, that distinction does not undermine Patent Owner's showing because the data still show that plecanatide exhibited unexpectedly improved interconversion stability under the conditions tested. Second, Petitioner's suggestion that these conditions are somehow not relevant is at odds with the fact that the data were apparently submitted to the FDA as part of Patent Owner's NDA. Ex. 2072 ¶ 13.

Petitioner's argument that Patent Owner should have additionally provided NMR data to confirm the structure of the peptide in each HPLC peak because "its HPLC protocol may have been optimized to separate

topoisomers of [uroguanylin] but not SP-304” is not persuasive. Reply 28 (citing Ex. 1063 ¶¶ 146–147). RP-HPLC was a known technique for separating topoisomers of uroguanylin. Ex. 2010, 230–31; Ex. 2011, 27–28; Ex. 2020, 223; Ex. 2028, TRUL00018269; Ex. 1063 ¶ 40. Thus, we see no reason why Patent Owner’s HPLC data are not themselves sufficient to show the alleged results. Indeed, the premise for Petitioner’s argument, i.e., Dr. Peterson’s testimony that the SP-304 peak “is relatively broad with some degree of tailing, raising the question of whether the HPLC protocol employed was simply lacking in optimization to resolve the topoisomers of this peptide” (Ex. 1063 ¶¶ 146–147), is just speculation. In contrast, we find the testimony Patent Owner offers from Dr. Shailubhai—who “directed the preparation and testing” resulting in Patent Owner’s data, provides detailed information regarding the HPLC conditions, and concludes that the chromatograph shows “only a single peak” for SP-304 (Ex. 2066 ¶¶ 17–20)—is more credible than the speculative testimony from Dr. Peterson that Petitioner cites and more consistent with the chromatographic evidence itself.

ii. Potency

Patent Owner argues that “Plecanatide is also unexpectedly more potent than human uroguanylin, stimulating significantly more cGMP production in T84 cells.” Resp. 60. In addition to Table 4 in the Specification, Patent Owner relies on the following data from Table 1 in Exhibit 2027. *Id.* at 61 (reproducing Ex. 2027, TRUL00018222 (Table 1)); *see also* Ex. 1001, 16:1–18 (Table 4) (showing some of the same results).

Table 1. Effects of SP-304, Uroguanylin, and Other Test Peptides in the T84 cGMP Stimulation Bioassay

Test Material	Concentration		cGMP Levels (pmol/well) *	EC ₅₀ **	
	Molar	ng/mL		Molar	ng/mL
SP-301 (uroguanylin)	0	0	0	10 ⁻⁶ M 2.3x10 ⁻⁷ M	1668 383.6
	10 ⁻⁹ M	1.668	0		
	10 ⁻⁸ M	16.668	12		
	10 ⁻⁷ M	166.8	82		
	10 ⁻⁶ M	1668	205		
	10 ⁻⁵ M	16680	254		
SP-302	0	0	0	10 ⁻⁶ M 3.5x10 ⁻⁷ M	1696 593.6
	10 ⁻⁹ M	1.696	0		
	10 ⁻⁸ M	16.96	8		
	10 ⁻⁷ M	169.6	62		
	10 ⁻⁶ M	1696	185		
	10 ⁻⁵ M	16960	248		
SP-303	0	0	0	10 ⁻⁶ M 2.4x10 ⁻⁷ M	1682 403.7
	10 ⁻⁹ M	1.682	0		
	10 ⁻⁸ M	16.82	12		
	10 ⁻⁷ M	168.2	82		
	10 ⁻⁶ M	1682	195		
	10 ⁻⁵ M	16820	254		
SP-304	0	0	0	10 ⁻⁷ M 1.1x10 ⁻⁷ M	1682 185.0
	10 ⁻⁹ M	1.682	0		
	10 ⁻⁸ M	16.82	17		
	10 ⁻⁷ M	168.2	149		
	10 ⁻⁶ M	1682	320		
	10 ⁻⁵ M	16820	315		

* cGMP levels in T84 cells after a 30-minute incubation.

** EC₅₀: median effective concentration (required to induce a 50% effect)

Table 1 above shows data for plecanatide (SP-304), human uroguanylin (SP-301) and other test peptides (SP-302 and SP-303) in the T84 cGMP stimulation bioassay. According to Patent Owner, these data show that “plecanatide increased cGMP production by about 56% at 10⁻⁶ M (320 v. 205 pmol/well)” and “achieved 50% of its maximal effect at less than half of the concentration of human uroguanylin (1.x10⁻⁷ M v. 2.3x10⁻⁷ M).” *Id.* at 60.

Patent Owner also relies on the data in Figure 2 of Exhibit 2028, reproduced below, showing potency of human uroguanylin, SP-302, SP-303, and SP-304 at pH 5.5, 6.0, 6.5, and 7.0. Resp. 61–62 (citing Ex. 2028, TRUL00018270); *see also* Ex. 2028, TRUL00018273 (showing raw data).

Figure 2: Effect of 30 Minute Incubations of 0.1 μ M Concentrations of SP-304, Uroguanylin, SP-302 and SP-303 on cGMP Production at Various Physiological pH Values in T84 Cells

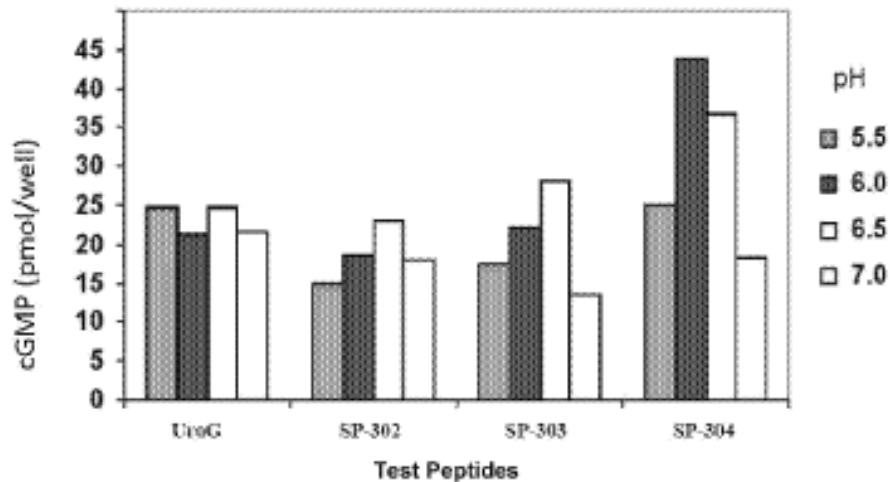


Figure 2 above is a bar graph of cGMP data for plecanatide, human uroguanylin, SP-302, and SP-303 at different pHs ranging from 5.5 to 7.0. According to Patent Owner, these data show “as compared to human uroguanylin, plecanatide increased cGMP production by about 105% at pH 6.0 and by about 49% at pH 6.5.” Resp. 61–62. Relying on testimony from Dr. Waldman, Patent Owner argues that this “increased pH sensitivity and better potency in the acidic portion of the intestine allows higher fluid secretion to treat constipation while its activity advantageously drops at pH 7 (in the colon) to minimize the likelihood of diarrhea.” *Id.* at 63 (citing Ex. 2025 ¶¶ 103–104, 110, 111).

Petitioner replies that the “magnitude” of the differences in Patent Owner’s potency data “is well within the . . . experimental error for these experiments and does not demonstrate any real, significant, or material difference in cGMP activity between the two peptides.” Reply 24 (citing Ex. 1063 ¶¶ 118–135). Referring to Table 4 in the Specification, Petitioner urges that its “cGMP values cannot be read to conclude a statistically

significant difference exists between the tested peptides, as even Dr. Shailubhai confirmed.” *Id.* (citing Ex. 1061, 33:19–34:15). Petitioner also asserts that “Dr. Davies testified a 40% difference in cGMP values for the same peptide in the same experiment does not reflect an actual difference but merely ‘a different way to interpret the data.’” *Id.* at 24–25 (quoting and citing Ex. 1060, 57:8–10, 54:3–4, 61:8–18, 74:8–17). Petitioner further urges that the raw data in Tables 2–3 of Exhibit 2028 “show **higher** nominal cGMP values **for uroguanylin** than for plecanatide at the same concentration with the same assay protocol.” *Id.* at 25 (citing Ex. 1063 ¶¶ 131–134). Moreover, Petitioner points to other data Patent Owner submitted to FDA, which it contends “indicates both peptides induced similar cGMP-production levels at the same concentration used to generate the Table 1 data [in Exhibit 2027] that was included in Table 4 [of the Specification].” *Id.* at 26 (citing Ex. 1069, 26, 30).

We find Patent Owner’s showing of unexpectedly improved potency and pH sensitivity persuasive. Patent Owner’s experiments show “the stimulation of cGMP synthesis in T84 cells by plecanatide (SP-304), human uroguanylin (SP-301),” as well as SP-302 and SP-303, as a measure of these peptides’ relative potency. *See* Ex. 2024 ¶¶ 217, 222, 224. Both the data in Table 4 of the Specification and Table 1 of Exhibit 2027 show that SP-304 produced a higher cGMP level in the T84 cell bioassay than SP-301 (uroguanylin). *See* Ex. 1001, 16:1–18 (Table 4); Ex. 2027, TRUL00018222 (Table 1). As Patent Owner points out, the difference at 10^{-6} M was more than 50% higher. *Id.* Moreover, Table 1 shows that SP-304 produced higher cGMP over a wide range of tested concentrations (10^{-7} M to 10^{-5} M) and that SP-304’s EC_{50} , i.e., the “median effective concentration (required to induce

a 50% effect),” was about half that of uroguanylin.²⁸ Ex. 2027, TRUL00018222 (Table 1). The data also show that SP-302 and SP-303 produced values similar to or slightly worse than uroguanylin at 10^{-6} M. *See* Ex. 1001, 16:1–18 (Table 4); Ex. 2027, TRUL00018222 (Table 1). Thus, Patent Owner’s data show that SP-304 exhibited improved potency values compared to both the parent peptide, human uroguanylin, as well as two other test peptides (SP-302 and SP-303) comprising different substitutions of the aspartic acid residues at positions 2 and 3.

The data in Figure 2 of Exhibit 2028 further support Patent Owner’s position. As Patent Owner points out, the data show that the Glu³ substitution resulted in increased cGMP production at pH 6 and 6.5 as compared to uroguanylin and the other tested peptides. Resp. 61–62 (citing Ex. 2028, TRUL00018270, TRUL000018273). Dr. Waldman offers testimony, which we credit, that both the increased activity and “pH sensitivity” are unexpected and “clinically meaningful” because

this unexpected pH sensitivity in the acidic areas of the intestines allows plecanatide’s unexpectedly superior selective activity in the [pH environment of] the small intestine to provide fluid into the intestine to treat constipation, rather than in the colon, where its activity drops (pH 7) and is thus less likely to cause adverse effects, like diarrhea. In other words, plecanatide’s targeted activity in the small intestine (not in the

²⁸ Petitioner asserts that the EC₅₀ values in Table 1 were amended to “eliminate[] the purported 10-fold potency difference” shown in the original version of that table. Reply 24. While this is true, Petitioner’s observation is inapposite because Patent Owner relies only on the amended values. *See* Resp. 60–61 (showing Table 1 as amended). Moreover, as Patent Owner explains, this amendment did not change the underlying cGMP data used to calculate those EC₅₀ values. Sur-reply (citing Ex. 2027, TRUL00018222).

colon) advantageously reduces the diarrhea rate in patient populations.

Ex. 2025 ¶ 110; *see also* Ex. 2025 ¶¶ 37–38 (explaining that “[a]s of 2002, ***all*** available treatment options [for constipation] targeted the colon or both the colon and small intestine,” which “is in stark contrast to plecanatide, which even today, is the only [chronic idiopathic constipation] and [irritable bowel syndrome with constipation] treatment to preferentially target the small intestine”). These differences, particularly the improved pH sensitivity at pH 6.0 and 6.5 shown in Figure 2 of Exhibit 2028, represent a difference in kind from the data for human uroguanylin.

Dr. Davies’s testimony that these results would have been unexpected because they show “that similar and/or seemingly minor modifications to human uroguanylin result in significant changes to biological activity in unpredictable ways” is persuasive. *See* Ex. 2024 ¶¶ 228–230. Li identifies both of the aspartic acid residues at position 2 and 3 in human uroguanylin as being of “particular interest.” *See* Ex. 1006, 52 (Fig. 6a), 54. And Petitioner appears to acknowledge that, by its own logic, a POSA relying on Li would also have been motivated to make the Glu² substitution. *See* Pet. 39 (“a POSA could also make [Glu²]-human uroguanylin as an operable synthetic uroguanylin”). However, Patent Owner’s data show that surprisingly *only* the Glu³ substitution results in improved potency compared to uroguanylin, whereas other conservative substitutions (i.e., the Glu² in SP-303 and the combination of the Glu² and Glu³ substitutions in SP-302) resulted in peptides with cGMP and EC₅₀ values similar to or worse than the parent peptide. Nothing in the record suggests such results would have been expected.

Petitioner’s argument that the differences between the values in Patent Owner’s data are immaterial and can be attributed to experimental error is unpersuasive. The data Patent Owner cites consistently show higher activity for SP-304 as compared to the other peptides tested. Moreover, Table 4 in the Specification states that the result for SP-304 is “statistically significant.” Ex. 1001, 16:19.²⁹ Petitioner asserts that Dr. Shailubhai “confirmed” that result was not statistically significant in his deposition. Reply 24 (citing Ex. 1061, 33:19–34:15). But all Dr. Shailubhai said was that the P value was calculated by someone else on his team and that he could not recall what SP-304 was compared to for that calculation. Ex. 1061, 33:19–34:15. None of that suggests the SP-304 result in Table 4 is not, in fact, significant.

So too, the testimony Petitioner cites from its cross-examination of Dr. Davies is inapposite and does not undermine Patent Owner’s showing. *See id.* at 24–25 (quoting and citing Ex. 1060, 57:8–10, 54:3–4, 61:8–18, 74:8–17). That testimony relates to a discrepancy between the value reported for SP-302 in Table 4 (225 pmol/well) and the value reported for SP-302 at a 10^{-6} M concentration in Table 1 of Exhibit 2027 (185 pmol/well). Ex. 1060, 70:6–71:5. Both SP-302 values are still well below the cGMP value reported for SP-304 (315 pmol/well) and we credit Dr. Davies’s testimony that looking at the data “as whole, it doesn’t change any interpretation you would put on the data.” *Id.* at 74:8–22.

²⁹ The note below Table 4 states “[t]he value observed for SP304 was statistically significant with a $p > 0.5$.” Ex. 1001, 16:19. Dr. Shailubhai explains—and Petitioner does not dispute—that there is a typographical error with the p value and that it “should say $p < 0.05$.” Ex. 2066 ¶ 9.

Dr. Peterson's testimony that a difference in the results for SP-304 in Patent Owner's data can be attributed to experimental error is not persuasive. *See* Ex. 1063 ¶¶ 118–35. Patent Owner offers multiple data sets, showing improved potency values for SP-304 relative to uroguanylin, SP-302, and SP-303. Ex. 1001, 16:1–18 (Table 4); Ex. 2027, TRUL00018222 (Table 1); Ex. 2028, TRUL00018270, TRUL00018273 (Table 2). In response, Dr. Peterson suggests that other statistical analyses could have been done or that these experiments should have been replicated to confirm their reliability. *See* Ex. 1063 ¶¶ 122–23. But that does not undermine the veracity of the data Patent Owner has presented.³⁰ Dr. Peterson also cites the same deposition testimony of Dr. Shailubhai and Dr. Davies discussed above, picking at the edges of Patent Owner's evidence by comparing results from different experiments in an attempt to suggest that any difference between the values for uroguanylin and SP-304 is unreliable. *See* Ex. 1063 ¶¶ 125–135. For example, Petitioner and Dr. Peterson point to the raw data for Patent Owner's pH sensitivity assays in Table 2 and compare them to the data for the thermostability assays in Table 3 of Exhibit 2028. *See* Reply 25 (citing Ex. 1063 ¶¶ 131–34; Ex. 2028, TRUL00018273 (Tables 2–3)). But these are different experiments and Petitioner has not sufficiently shown that the conditions (e.g., pH) are similar such that the comparison Dr. Peterson makes is accurate.

For these reasons, we credit the testimony of Dr. Davies and Dr. Waldman that Patent Owner's data demonstrate significant and

³⁰ Moreover, at least for the results in Table 1 the “[s]amples were run in duplicate[] incubations and each sample was run as duplicates in ELISA test.” Ex. 2027, TRUL00018211; Ex. 2023 ¶ 15.

unexpectedly improved potency and pH sensitivity for SP-304 as compared to uroguanylin (*see* Ex. 2024 ¶¶ 216–30; Ex. 2025 ¶¶ 95–111) over the competing testimony Petitioner offers from its experts.

iii. Binding affinity

As discussed above, we find that Patent Owner’s own experiments (i.e., the results in Table 4 of the Specification and in Exhibits 2027 and 2028) are sufficient to demonstrate unexpected results of the claimed peptide, but there is additional work from an external laboratory that bolsters this finding. Specifically, Patent Owner asserts that “[p]lecanatide also has an unexpected and superior binding affinity as compared to human uroguanylin.” Resp. 66. Patent Owner’s binding affinity results differ from the other categories of results because they are based on data from experiments described in a third-party, post-issuance research article, Liu (Exhibit 2046), rather than the data in the Specification of the ’786 patent and Exhibits 2027 and 2028. Before turning to what Liu’s data show, we first address the parties’ arguments regarding whether Liu’s binding affinity experiments tested the closest prior art.

Patent Owner asserts that “Liu used radio-labelled human uroguanylin and plecanatide, purified to their topoisomers, to evaluate their relative binding affinities.” PO Remand Br. 14 (citing Ex. 2046, 3777 (Abstr.), 3778 (Materials and Methods), 3779)). According to Patent Owner, “[b]efore testing binding affinities, Liu states that he obtained and purified to ‘homogeneity by RP-HPLC’ the active forms of human uroguanylin and plecanatide (referred to as ‘E3-uroguanylin’), which he confirmed through the use of the T84 cell bioassay.” *Id.* (quoting Ex. 2046, 3778). Patent

Owner further asserts that the RP-HPLC chromatogram in Figure 3 of Liu confirms this understanding. *Id.* at 15.

Petitioner argues “there is no evidence that the purified active form of uroguanylin was used in Liu” because while “Patent Owner points to the ‘chromatogram for radiolabeled peptides [in Figure 3], showing singular peaks for human uroguanylin and plecanatide,” it “makes no showing that the ‘singular peak’ for uroguanylin *actually* represents the active topoisomer.” Pet. Remand Resp. 25 (quoting PO Remand Br. 15)). In addition, Petitioner suggests Liu’s data is faulty because Liu provides no information regarding how the uroguanylin peptides were stored and how long the peptides were in solution prior to being tested.” *Id.* at 24–25 (citing Ex. 1078 ¶ 20; Ex. 1079, 136:8–19).³¹

Patent Owner replies, urging that “[i]t cannot be credibly argued that the peer-reviewed Liu publication . . . would have studied peptides with sizable inactive content” given that “[b]y the time Liu published in 2009, uroguanylin was well-known to interconvert,” “Liu states that he synthesized and purified” the uroguanylin “via RP-HPLC to homogeneity,” and Liu confirms the peptides “had comparable activity” in its “T84 cancer cell line test.” PO Remand Reply 9–10 (citing Ex. 2046, 3778–3780).

Patent Owner again has the better position. Liu plainly discloses that RP-HPLC was used to purify both uroguanylin and plecanatide (E³-uroguanylin) to homogeneity. Ex. 2046, 3778. Moreover, the

³¹ Petitioner also cites statements in a letter from the American Peptide Company that it contends describe the uroguanylin peptides used in Liu. Pet. Remand Br. 25 (citing Ex. 1067, 86). That letter is the subject of Patent Owner’s Motion to Exclude (Paper 101). As explained below, we grant that motion and thus do not consider Petitioner’s argument based on the same.

chromatograms in Figure 3 show a single, well-defined peak, which is consistent with the other evidence demonstrating topoisoimeric purity. *Id.* at 3780. It is true that the Figure 3 chromatograms show the radiolabeled versions of the peptides used in Liu’s experiments. *See* Ex. 1078 ¶ 21 (observing that the ¹¹¹In-DOTA-conjugated peptides represent a structural modification of the underlying peptide). But Liu separately uses precisely the same wording (“purified to homogeneity by RP-HPLC”) to describe purification of the unconjugated peptides. Ex. 2046, 3778. The evidence in Liu therefore more reasonably supports the position that the chromatograms in Figure 3 are representative of a level of “homogeneity” achieved through its purification of those peptides. *Id.*

Petitioner’s other arguments are premised on speculation that Liu tested the purified inactive form of human uroguanylin instead of the active topoisoimer. Such speculation is unfounded for several reasons. First, the foundation for Liu’s study is uroguanylin’s known activity as an “agonist that binds the guanylate cyclase C receptor (GC-C).” Ex. 2046, 3777 (Abstr.). That receptor “is overexpressed in human colorectal cancer.” *Id.* Liu’s study seeks to use “radiolabeled uroguanylin analogs for in vivo detection” of colorectal cancer by testing their binding affinity for GC-C and measuring in vivo tumor uptake of the radiolabeled peptides. *Id.* The use of the inactive topoisoimer would therefore undermine the purpose of Liu’s study.

Second, both the uroguanylin used in Liu’s binding assay and the radiolabeled conjugate in Liu’s in vivo testing exhibited activity that Liu was able to compare to the corresponding results for plecanatide. *See id.* at 3779–3780. While plecanatide had significantly higher binding affinity, the fact

that Liu's purified uroguanylin and uroguanylin conjugate also showed activity in these tests is inconsistent with Petitioner's speculation that it was the purified *inactive* topoisomer, rather than the active topoisomer, that Liu tested.

Third, given uroguanylin's known interconversion issues and the significant effort Liu describes taking to purify it via RP-HPLC, it seems unlikely that Liu would then store it or unduly delay its testing in solution in a manner that would allow otherwise avoidable interconversion back to the inactive topoisomer to occur.

In sum, Patent Owner has sufficiently demonstrated that Liu's experiments used and tested the closest prior art, i.e., the purified (>95%) active topoisomer of uroguanylin. We now turn to what Liu's data show.

As noted above, based on the data in Liu, Patent Owner asserts that “[p]lecanatide also has an unexpected and superior binding affinity as compared to human uroguanylin.” Resp. 66. According to Patent Owner, the data in Liu's Table 1, reproduced below, shows that plecanatide (E3-uroguanylin) binds to the GC-C receptor with 8-fold higher affinity than native human uroguanylin. Resp. 66 (reproducing table from Ex. 2046, 3779).

Table I. Calculated and observed (M+H)⁺ values and IC₅₀ values (± SD) for characterized peptides.

Peptide	(M+H) ⁺ Calc.	(M+H) ⁺ Obs.	IC ₅₀ (nM)
Uroguanylin	1667.6	1667.7	39.8±14.9
DOTA-uroguanylin	2053.6	2053.9	34.5±3.3
E ³ -uroguanylin	1681.6	1681.6	5.0±0.3
DOTA-E ³ -uroguanylin	2067.6	2067.9	9.6±2.9

Table 1 above shows IC₅₀ values³² from a competitive binding assay described in Liu. *See* Ex. 2046, 3778–3779. Patent Owner compares the IC₅₀ value reported for uroguanylin (39.8 ± 14.9 nM) to the IC₅₀ value for E³-uroguanylin (5.0 ± 0.3 nM) from Table 1 above. Relying on testimony from Dr. Davies and Dr. Waldman, Patent Owner argues that this approximately “8-fold increase in binding affinity would have been unexpected given the single methylene group difference [between aspartic acid and glutamic acid] and also meaningful, for it demonstrates substantially greater potency for cGMP production to treat constipation as shown above.” Resp. 66 (citing Ex. 2024 ¶ 243; Ex. 2025 ¶¶ 94, 110–11).

Petitioner does not dispute that Liu’s data show an 8-fold increase in binding affinity. *See* Reply 26. Instead, Petitioner urges this is not an unexpected result because Patent Owner “does not contend this binding-affinity increase translated into an **8-fold** potency increase.” *Id.* (citing Ex. 1064 ¶¶ 47–49).

We disagree with Petitioner. The data in Liu’s Table 1 shows that plecanatide exhibited significantly improved binding affinity over uroguanylin. Liu observes that “[a] conservative D3E substitution [i.e., the Glu³ substitution] significantly increased the binding affinit[y] of the resultant uroguanylin analogs.” Ex. 2046, 3779. This observation supports the testimony of Patent Owner’s experts, which we credit, that a POSA

³² As Dr. Davies explains, “IC₅₀ is a quantitative measure that indicates how much (or what concentration) of a particular inhibitory substance (e.g. drug) is needed to inhibit, in vitro, a given biological process or biological component by 50%.” Ex. 2024 ¶ 239; *see also* Ex. 2025 ¶ 90 (similar explanation from Dr. Waldman).

would not have expected this conservative substitution to result in such a large improvement in binding affinity. Ex. 2024 ¶ 243; Ex. 2025 ¶ 94.

The record also shows that binding affinity is a relevant consideration because it is an indicator of potency. Dr. Davies testifies that plecanatide’s “unexpectedly better” IC₅₀ “is indicative of the superior activity of plecanatide.” Ex. 2024 ¶¶ 241–43. Dr. Waldman similarly testifies that this result is “significant because plecanatide would be seen as more potent in activating the receptors to stimulate greater cGMP production and potentially better relief of constipation.” Ex. 2025 ¶ 94. Their testimony is consistent with the use of binding affinity assays to analyze and compare similar peptides in various other references. *See, e.g.*, Ex. 1005, Fig. 3B, 6:15–22; Ex. 1021, 2709; Ex. 2065, F345F (using binding assays to analyze biological activity and relative potency of uroguanylin and guanylin).

Petitioner’s argument that Patent Owner’s showing is insufficient unless the difference in binding affinity translates into a corresponding potency increase is not credible given Petitioner’s own reliance on such data in its *prima facie* case. *See* Reply 26. As Patent Owner points out, Currie uses a similar competitive binding assay to assess the “relative affinity” of human uroguanylin, human guanylin, and STa. *See* Paper 71 (“Sur-reply”), 25; Ex. 1005, Fig. 3B, 6:15–22. Petitioner relies on this data, reproducing the binding affinity results shown in Currie’s Figure 3B in the Petition and urging that “Figure 3B . . . illustrates the increased potency of human uroguanylin compared to human guanylin.” Pet. 23. So too, Li cites uroguanylin’s “10-fold higher . . . affinity” as support for its teaching identifying certain positions of “particular interest.” *See* Ex. 1006, 54. This

is the same teaching Petitioner relies upon to support its conservative substitution rationale for modifying uroguanylin. *See* Dec. 20–38 (§ III.E.ii).

The testimony Petitioner cites from Dr. Epstein does not show otherwise. According to Dr. Epstein, the result in Liu “would not have been viewed as ‘superior’ or ‘unexpected’ from a clinical standpoint” because “[b]inding affinity is really only relevant to a clinician to the extent it translates to activity.” Ex. 1064 ¶ 48. To the extent Dr. Epstein is suggesting that a POSA would not consider significantly improved binding affinity relevant, his testimony is belied by Currie, Li, and the other evidence noted above. Accordingly, we do not find the testimony Petitioner cites (i.e., Ex. 1064 ¶¶ 47–49) credible to rebut Patent Owner’s showing of significant and unexpectedly improved binding affinity.

D. Petitioner’s argument that the results are not reasonably commensurate

In the Decision, we found that Patent Owner’s evidence of unexpected results was reasonably commensurate with the scope of claim 1 even if that claim encompassed both the active and inactive topoisomer of the recited peptide. Dec. 49. The Federal Circuit did not reach Petitioner’s challenge to that finding on appeal. CAFC Dec. 9. On remand, Petitioner reiterates its position that because the challenged claims “encompass both an ‘active’ and ‘inactive’ form of plecanatide,” Patent Owner has not shown that its results are commensurate with the scope of the claims. Pet. Remand Resp. 28–29; *see* Reply 22–23 (prior argument). We disagree.

Claim 1 is narrowly drawn to a peptide consisting of a single 16 amino acid long sequence. The other challenged claims are likewise limited to a peptide consisting of that sequence. *See* Ex. 1001, 37:4–38:9. While that

peptide is subject to topoiosomerism, both topoisomers are still the same compound with the same chemical structure. Ex. 2024 ¶ 50 (“[A] ‘topoisomer’ is the same chemical compound having different spatial arrangement of the various atoms in the molecule that occur because of rotations around bonds.”); *see also* CAFC Oral Arg. at 9:54–9:59, *available at* https://cafc.uscourts.gov/oral-arguments/24-1011_04072025.mp3 (Petitioner’s counsel acknowledging that uroguanylin’s topoisomers “both have the same structure”). Thus, claim 1 is limited to just one compound, plecanatide.

The experiments Patent Owner relies upon test plecanatide against the nearly identical structure in human uroguanylin under biologically-relevant conditions where both peptides have been purified to an equal fraction (>95%) of the active topoisomer. These experiments show significant differences in potency and binding affinity resulting from the addition of a single methylene group in the side chain (i.e., glutamate versus aspartate) at position 3. We agree with Patent Owner that there is no reason to believe that similar results would not be obtained if the plecanatide and human uroguanylin were tested as mixtures containing higher fractions of their inactive topoisomers. *See* PO Remand Br. 2–3, 9–10 (“proportionally decreasing the amount of active topoisomer, and thus increasing the amount of inactive topoisomer, would effectively add a null set to the biological activity already observed in the best-on-best comparison”); Ex. 2077, 21:14–21 (Petitioner’s expert testifying that the inactive topoisomers of uroguanylin and plecanatide have similarly, low activity). This is consistent with Marx’s observation, based on experimental data therein, that the amount of cGMP produced by mixtures of uroguanylin with different ratios

of active to inactive topoisomers “were between the pure isoforms A and B, with a linear relation to the molar ratio of uroguanylin isomers” and thus “an antagonistic effect” of the inactive topoisomer “can be excluded.” Ex. 2010, 238. Moreover, Patent Owner’s data showing that plecanatide is less subject to interconversion applies to both its active and inactive topoisomeric forms. Ex. 2028, TRUL00018269.

The mere fact that Patent Owner’s experiments did not also test peptide mixtures comprising higher fractions of the inactive topoisomer does not defeat Patent Owner’s unexpected results showing. “[A]bsolute identity of scope’ is not required” for the claims to be reasonably commensurate in scope with the experimental data. *Genetics Institute, LLC v. Novartis Vaccines and Diagnostics, Inc.*, 655 F.3d 1291, 1308–09 (Fed. Cir. 2011). Nor is Patent Owner “required to test every embodiment within the scope of his or her claims” where, as here, there is adequate basis to conclude that “other embodiments falling within the claim will behave in the same manner.” *In re Kao*, 639 F.3d 1057, 1068 (Fed. Cir. 2011).

Finally, to the extent Petitioner suggests that Patent Owner’s results are not commensurate because they do not show that plecanatide’s *inactive* topoisomer exhibits superior properties in comparison to uroguanylin’s *active* topoisomer, we disagree. *See* Reply 22–23 (arguing that because Patent Owner told the EPO³³ “the difference in active and other topoisomers was at least 7-fold,” which is less than the difference distinguishing the prior art, Patent Owner cannot show that its results are commensurate in scope

³³ Citing Ex. 1067, 102 (“It is very well-known in the literature that the potency of the A-form of uroguanylin is at least 7-fold higher than the potency of the B-form of uroguanylin.”).

with the claims). Patent Owner does not need to show that plecanatide excels over uroguanylin “in all common properties.” *In re Chupp*, 816 F.2d 643, 646 (Fed. Cir. 1987); *see also Novo Nordisk A/S v. Caraco Pharm. Labs., Ltd.*, 719 F.3d 1346, 1366 (Fed. Cir. 2013) (explaining it was enough that the patent owner demonstrated unexpected results in a particular patient population, it was not necessary to show those results in other populations).

Indeed, the facts in *Chupp* are illustrative. Like the claims here, the claims in *Chupp* were limited to one chemical compound that differed from the closest prior art by a single methylene group. *Chupp*, 816 F.2d at 644. The applicant submitted evidence showing that its claimed compound, which functioned as a selective herbicide, demonstrated superior results when used with certain crops, but not others. *Id.* at 645. The Board of Patent Appeals found those results were not reasonably commensurate “because the claims were limited to no particular weed or crop.” *Id.* But the Federal Circuit reversed, explaining that it was not necessary to demonstrate that the claimed compound “produce[s] superior results in every environment in which the compound may be used” because the evidence demonstrating superior activity in at least some environments was sufficient to rebut the prima facie case of obviousness. *Id.* at 646–47. So too here, given the narrow scope of the claims, we find that it is enough that Patent Owner has shown that plecanatide exhibits superior results in the tested environment (i.e., solution-based assays of purified (>95%) active topoisomer uroguanylin versus plecanatide of the same purity), particularly as those experiments were designed to obtain information as to how the peptides might behave *in vivo* in the gastrointestinal tract. Ex. 2072 ¶ 11.

In sum, we find that there is a sufficient nexus between Patent Owner’s results and the challenged claims and that those results are reasonably commensurate with the same. Accordingly, for all of reasons explained above, the unexpected results Patent Owner relies upon are probative of non-obviousness.

E. Conclusion for Ground 1

Weighing the record as a whole in view of the findings and analysis above as well as our findings and analysis relating to Petitioner’s prima facie showing in the Decision, we conclude that Petitioner has not met its burden to show that claim 1 would have been obvious over Currie and Li. The evidence suggesting that it would have been obvious to make the Glu³ substitution because it is a conservative substitution (*see* Dec. 20–38 (§ III.E.ii)) is substantially outweighed by Patent Owner’s comparatively strong evidence demonstrating unexpected results relating to improved interconversion stability, potency, and binding affinity from the Glu³ substitution. *See Millennium Pharms., Inc. v. Sandoz Inc.*, 862 F.3d 1356, 1368 (Fed. Cir. 2017) (“Evidence of objective indicia [such as unexpected results] ‘can be the most probative evidence of nonobviousness in the record,’ and . . . enable[s] ‘the court to avert the trap of hindsight.’”) (quoting *Leo Pharm. Prods., Ltd. v. Rea*, 726 F.3d 1346, 1358 (Fed. Cir. 2013)). Indeed, the interconversion and potency categories of Patent Owner’s unexpected results showing are particularly strong such that even without the additional results in Liu, we would still reach the same ultimate conclusion on obviousness. Accordingly, Petitioner has not shown by a preponderance of the evidence that claim 1 is unpatentable.

F. Other Grounds

Petitioner's other grounds challenge claims 2–6 and are premised on Petitioner's showing in Ground 1 for the limitations of claim 1. *See* Pet. 40–53. Petitioner relies on the additional references asserted in these grounds only for their teachings regarding the additional limitations recited in the dependent claims. *Id.* Because Petitioner has not shown that claim 1 would have been obvious over Currie and Li, the premise for Petitioner's other grounds has not been established. Accordingly, Petitioner has not shown by a preponderance of the evidence that claims 2–6 are unpatentable.

IV. PATENT OWNER'S MOTION TO EXCLUDE

Patent Owner moves to exclude page 86 of Exhibit 1067 (the “American Peptide letter”) as hearsay. Paper 101, 1 (“MTE”). This motion is essentially a continuation of Patent Owner's earlier motion to exclude Exhibit 1067 (Paper 53), which we granted in part with respect to the materials submitted by the opponent in the EPO opposition. *See* Dec. 63–72 (referring to these materials as “the Currie submission” and excluding the same as hearsay). The American Peptide letter was apparently among the materials submitted by the EPO opponent. *See* Ex. 1067, 93 (“Opponent 1 files herewith a letter from the American Peptide Company”). However, at the time of our Decision, we did not discern Petitioner's reliance on that letter so page 86 of Exhibit 1067 was not included amongst the pages we excluded in the Currie submission. Dec. 63, 69 (excluding “pages 88–98 and 120–127 of Exhibit 1067,” but dismissing the remainder of Patent Owner's motion as moot).

On remand, Petitioner specifically relies on the American Peptide letter. Pet. Remand Resp. 25 (quoting Ex. 1067, 86). Patent Owner sought authorization to file the present motion to exclude, which we allowed. Paper 99. Petitioner opposes that motion. *See* Paper 102 (“MTE Opp.”).

Having considered both sides’ arguments, we grant Patent Owner’s motion. The American Peptide letter contains out of court statements from a third party submitted by the EPO opponent. The statements in the letter were not made under oath, nor is there evidence that Patent Owner was ever able to question the author of that letter or anyone else with personal knowledge of the statements therein. Petitioner argues that the American Peptide letter is evidence “to doubt that the ‘singular peak’ for uroguanylin in Liu represents the active topoisomer of uroguanylin” because it evidences that American Peptide “provided its customers with the ‘B’ (i.e., inactive) form of uroguanylin at the time.” Pet. Remand Resp. 25. That is, Petitioner relies on the American Peptide letter for the truth of the matters asserted therein. Such use plainly implicates the rule against hearsay and falls squarely within our prior reasoning for excluding the Currie submission portion of Exhibit 1067. We incorporate that analysis herein. *See* Dec. 61–72.

Petitioner asserts that the American Peptide letter is admissible under the residual hearsay exception in FRE 807. MTE Opp. 3–5. We disagree. The American Peptide letter was submitted by the EPO opponent specifically for the purpose of opposing a foreign counterpart to the ’786 patent. We see no reason to distinguish it from the other materials previously excluded as hearsay in the Currie submission. Petitioner’s assertion that it contains sufficient guarantees of trustworthiness because it was “provided by a neutral third-party” (American Peptide) and “an individual having first-

hand knowledge, [American Peptide Company's] VP of sales and marketing" is speculation. *See id.* at 3–4. Based on the record before us, Patent Owner has never been afforded an opportunity to examine anyone regarding the statements in this letter or even confirm that it is, in fact, a letter from the American Peptide Company. The mere fact that it was submitted in a foreign patent proceeding does not make it any more trustworthy than any other untested assertion offered by an adverse party in a legal proceeding.

For these reasons, we grant Patent Owner's motion to exclude page 86 of Exhibit 1067.

V. CONCLUSION

Petitioner has not shown, by a preponderance of the evidence, that claims 1–6 of the '786 patent are unpatentable.

Claims	35 U.S.C. §	Reference(s)/ Basis	Claims Shown Unpatentable	Claims Not shown Unpatentable
1	103(a)	Currie, Li		1
2, 4, 5	103(a)	Currie, Li, Narayani		2, 4, 5
3–5	103(a)	Currie, Li, Narayani, Campieri		3–5
6	103(a)	Currie, Li, Ekwuribe		6
Overall Outcome				1–6

VI. ORDER

Accordingly, it is:

ORDERED that Petitioner has not shown that claims 1–6 of U.S. Patent 7,041,786 B2 are unpatentable;

FURTHER ORDERED that Patent Owner’s Motion to Exclude (Paper 101) is granted; and

FURTHER ORDERED that, because this is a Final Written Decision, parties to this proceeding seeking judicial review of our Decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

IPR2022-00722
Patent 7,041,786 B2

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